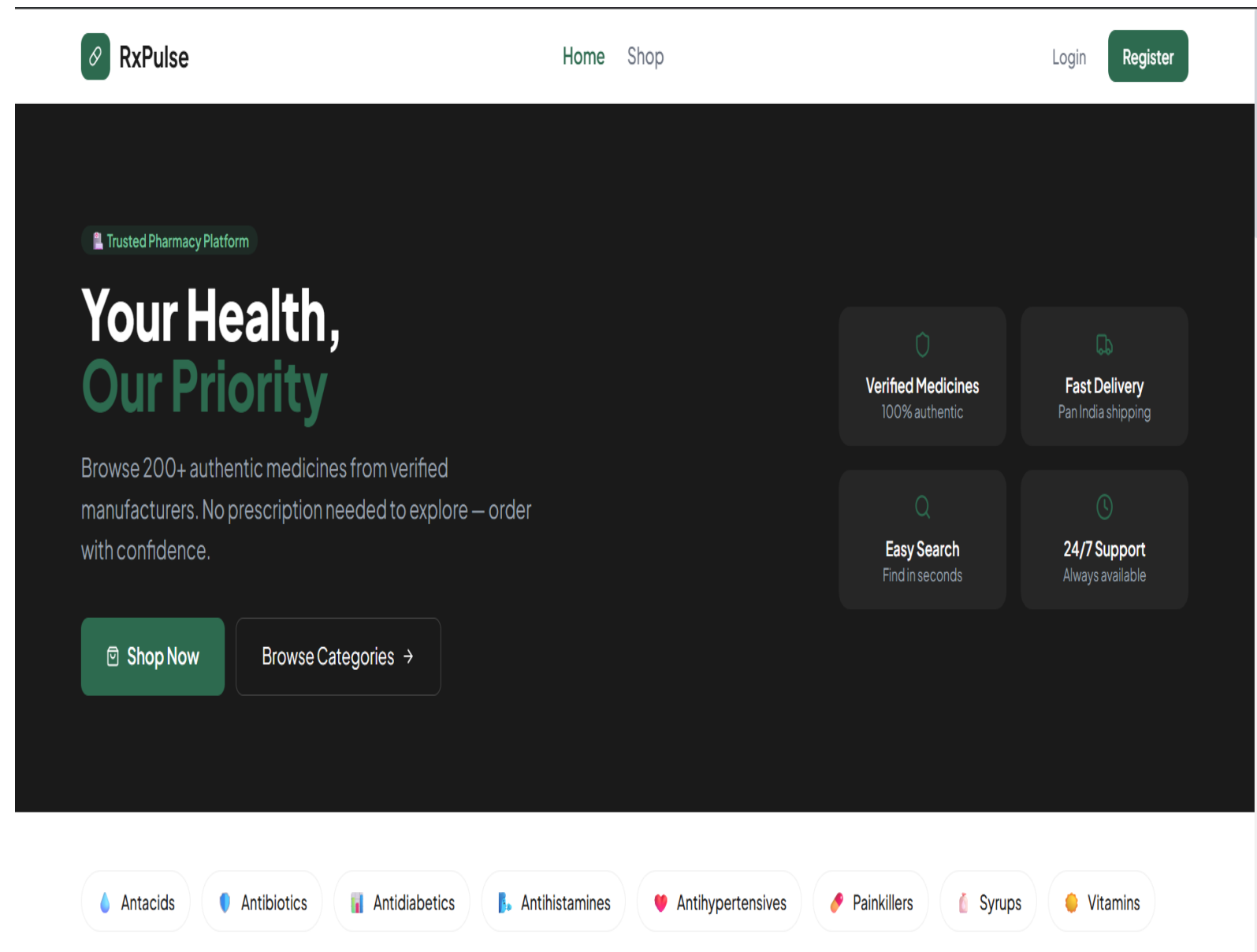


RxPulse — Pharmacy Management Platform

Cloud-native microservices with
DevOps & GitOps — built for reliability, security, and
zero-downtime deployments.



What is RxPulse?

RxPulse is a cloud-native pharmacy management platform — three backend microservices, a React + TypeScript frontend, and a DevOps pipeline built for production-grade reliability.

3 Backend Microservices

Each with an isolated MongoDB, deployed as independent Kubernetes workloads.

React + TypeScript Frontend

Served via Nginx with Argo Rollouts for blue-green, zero-downtime releases.

DevOps & GitOps Pipeline

GitHub Actions CI, ArgoCD GitOps, and security scanning at every stage.

Application Components

Three bounded-context microservices, each owning its data and communicating over the Kubernetes Gateway API.



User Service

Manages authentication and JWT tokens.

- Admin → Full system access (dashboard, inventory, movements, alerts)
- Customer → Browse & order medicines



Catalog Service

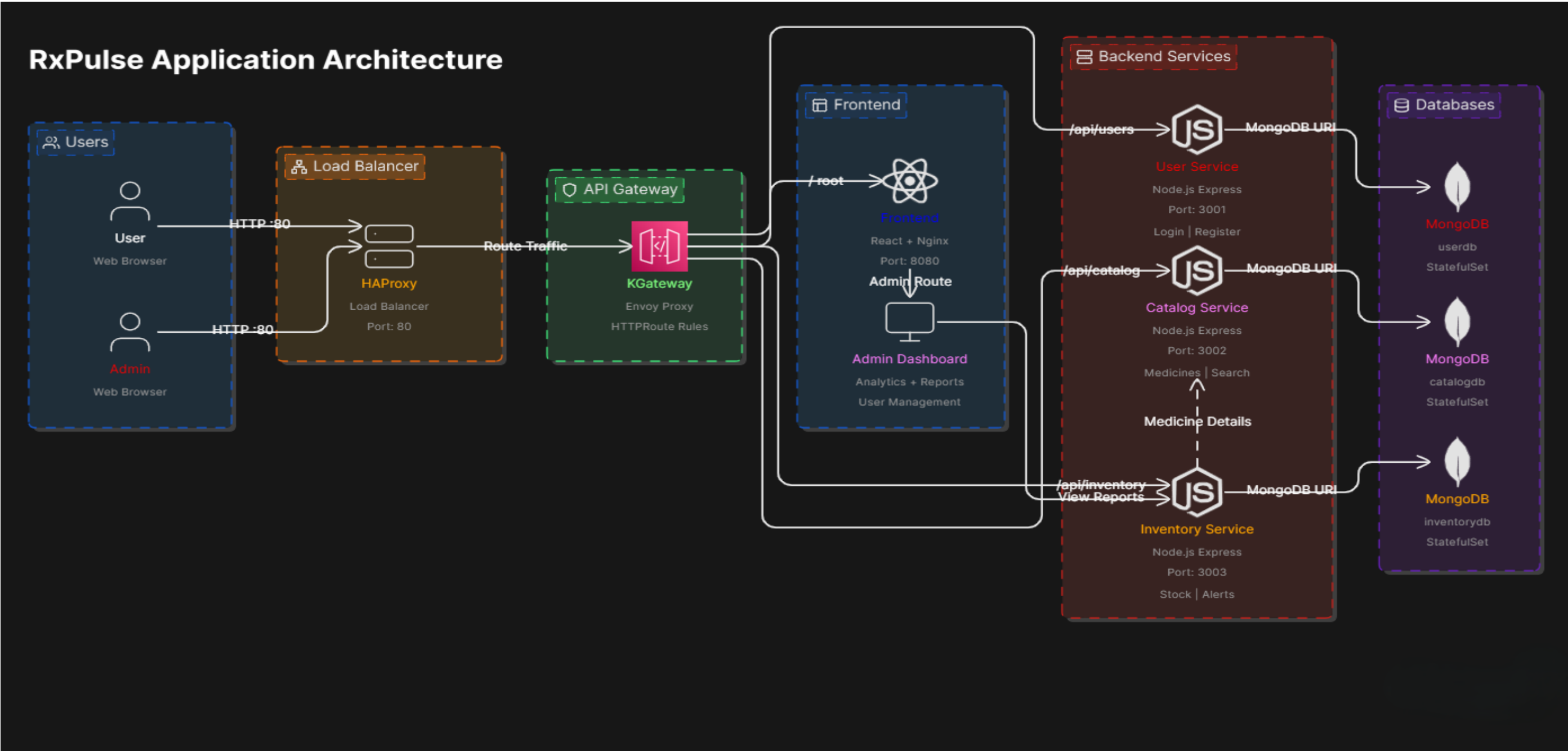
Manages medicine records, pricing, and availability. Exposes a REST API consumed by the frontend for product browsing and search.



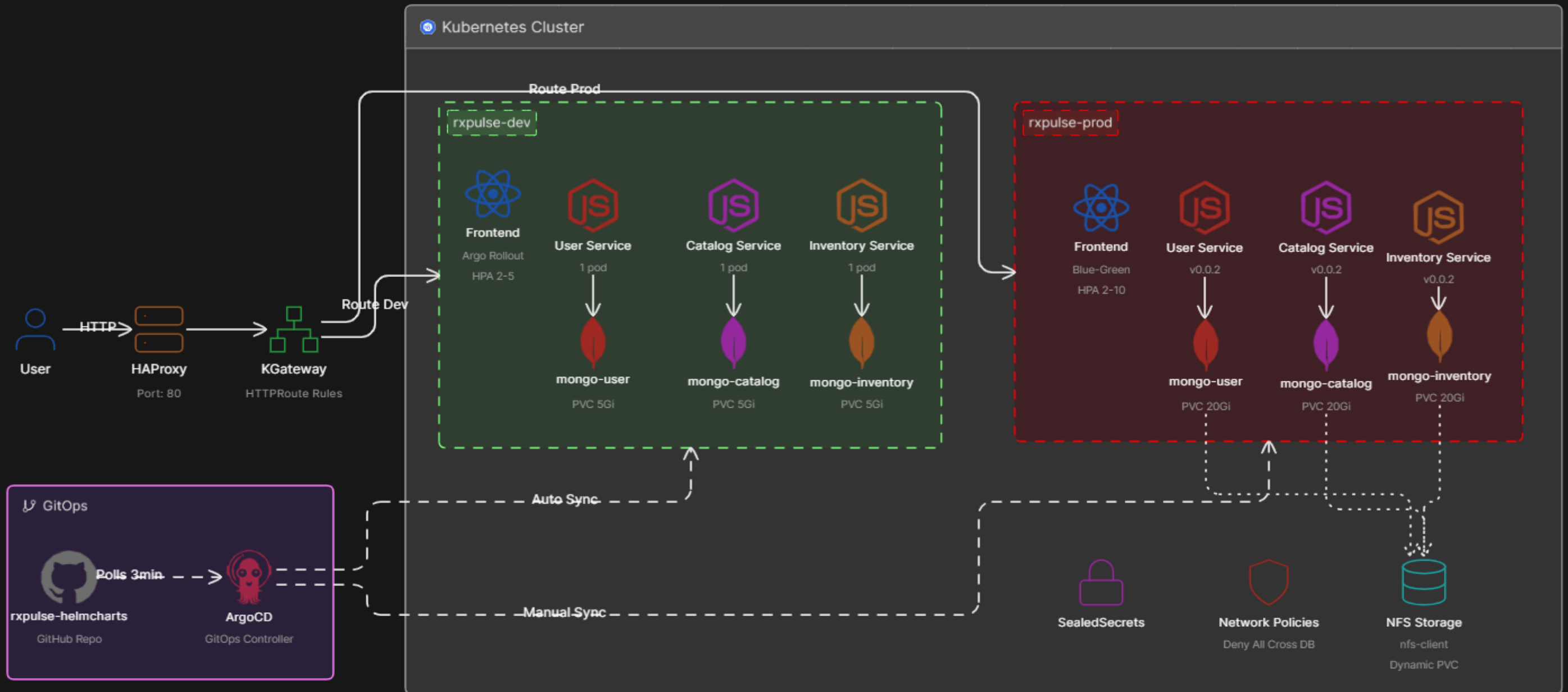
Inventory Service

Tracks stock levels in real time and triggers low-stock alerts. Maintains a dedicated MongoDB for high-frequency write operations.

Application Architecture



RxPulse Kubernetes & DevOps Architecture



Security Implementation — Defense in Depth



JWT-based authentication secures all APIs.

- Admin → Full access (dashboard, inventory, alerts)
- Customer → Browse medicines & place orders



NetworkPolicy

Default-deny-all posture. Each MongoDB accepts connections only from its owning service. Prevents lateral movement and data exfiltration.



SealedSecrets (Prod)

Sensitive data (DB passwords, API keys) encrypted at rest in Git. Only the prod cluster can decrypt using its sealing key.

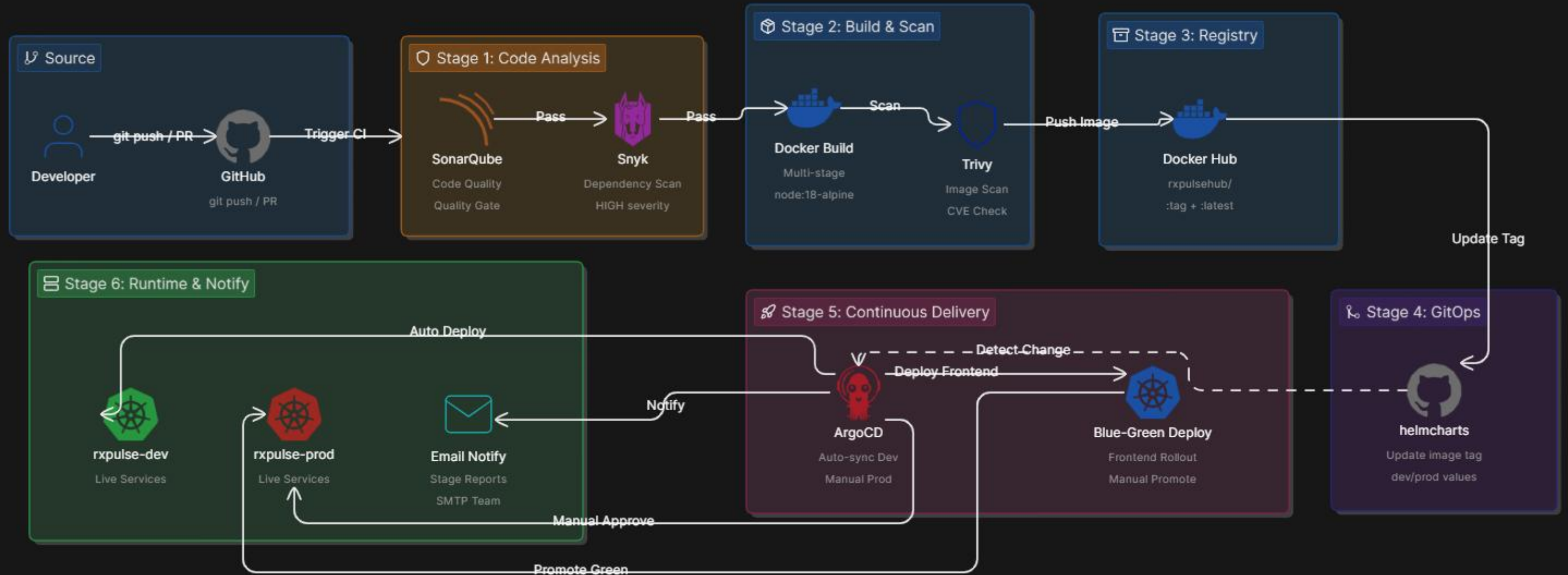


Non-Root Containers

All Docker images run as non-root users. Reduces blast radius if a container is compromised. Combined with read-only filesystems where possible.

Security is layered: identity (JWT), network isolation (NetworkPolicy), secrets management (SealedSecrets), and container hardening (non-root). No single point of failure.

RxPulse CI/CD Pipeline



GitOps with ArgoCD & Argo Rollouts

ArgoCD — GitOps Controller

A separate Helm repo (`rxpulse-helmcharts`) is the single source of truth. ArgoCD reconciles cluster state to match Git.

- **Dev:** Auto-sync on every commit
- **Prod:** Manual approval gate before sync

Argo Rollouts — Blue-Green

Frontend uses Argo Rollouts for zero-downtime releases.

- **Blue (Active):** Live production traffic
- **Green (Preview):** New version staged
- **Promote:** Manual switch after validation

GitOps ArgoCD

argo
v3.3.7

Applications

Settings

User Info

Documentation

Application filters

☐ ★ Favorites Only

SYNC STATUS

☐ ⏸ Unknown 0

☐ ✓ Synced 2

☐ ⚠ OutOfSync 0

HEALTH STATUS

☐ 🔄 Progressing 0

☐ ⏸ Suspended 0

☐ ❤️ Healthy 2

☐ ❤️ Degraded 0

☐ 👻 Missing 0

Applications

+ NEW APP

↻ SYNC APPS

🔄 REFRESH APPS

🔍 Search applications...

/

APPLICATIONS TILES

🗪

📋

📊

Log out

Sort: name ▼ Items per page: 10 ▼

rxpulse-dev

Project: rxpulse

Labels: environment=dev

Status: ❤️ Healthy ✓ Synced

Reposit... https://github.com/RxPulse/rxpulse-he...

Target ... develop

Destina... in-cluster

Names... rxpulse-dev

Create... 04/25/2026 19:32:01 (3 days ago)

Last Sy... 04/27/2026 16:59:25 (a day ago)

🔄 SYNC

🔄

✖

rxpulse-prod

Project: rxpulse

Labels: environment=prod

Status: ❤️ Healthy ✓ Synced

Reposit... https://github.com/RxPulse/rxpulse-he...

Target ... main

Destina... in-cluster

Names... rxpulse-prod

Create... 04/25/2026 22:00:31 (3 days ago)

Last Sy... 04/27/2026 10:44:12 (a day ago)

🔄 SYNC

🔄

✖

Argo Rollouts



argo

Rollouts v1.9.0+838d4e7



NAMESPACE

rxpulse-prod

frontend



! Abort

Promote

PromoteFull

Summary

Strategy



Containers

 Edit

frontend

rxpulsehub/rxpulse-frontend:v0.1.0



Revisions

Revision 15

rxpulsehub/rxpulse-frontend:v0.1.0

frontend-7c56b678dc



👍 stable

 active



Revision 14

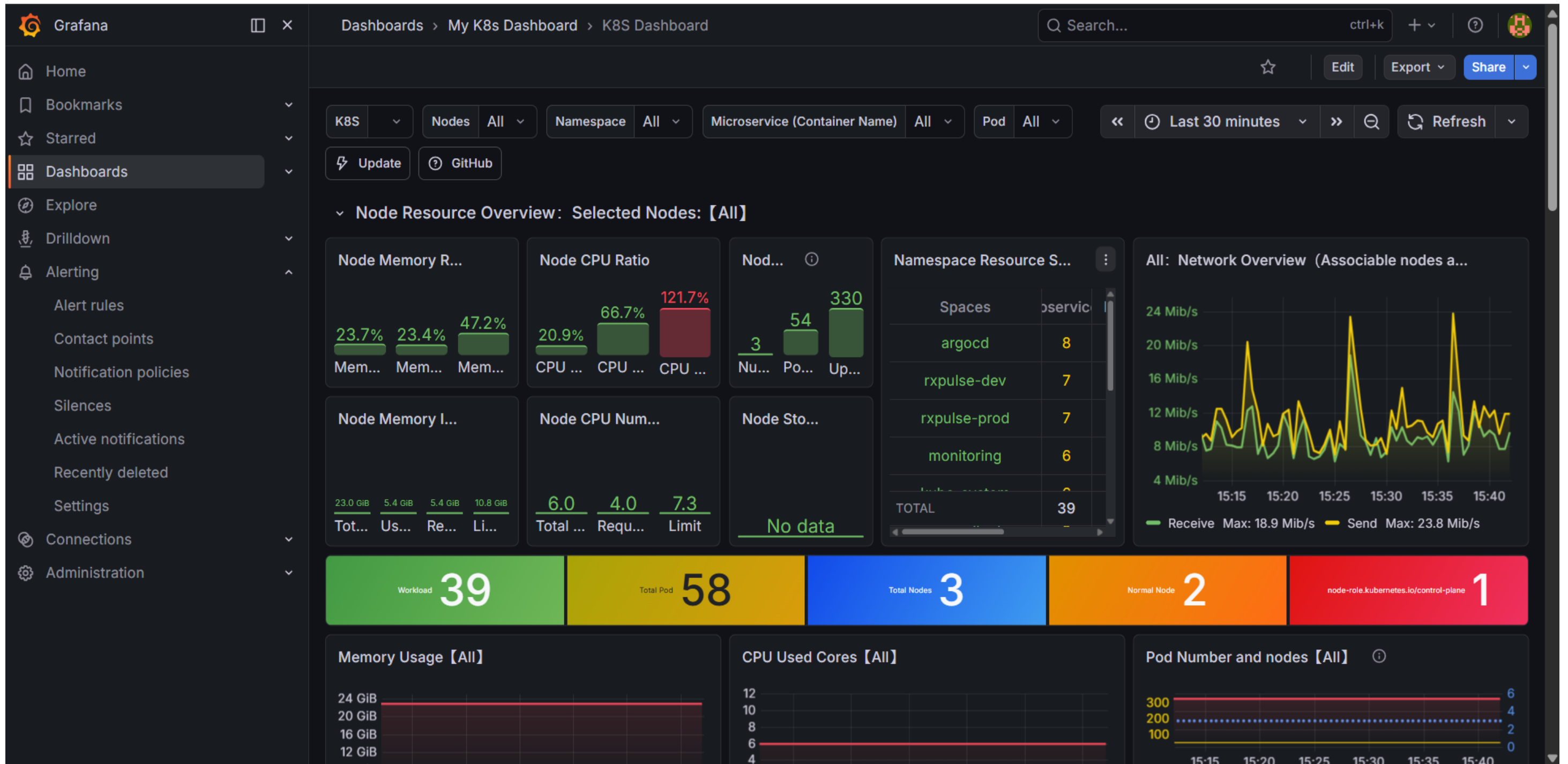
rxpulsehub/rxpulse-frontend:v0.2.0

frontend-7bcf6966b6

No Pods!

 Rollback

Monitoring – Grafana



Alerts

 Grafana

Home

Bookmarks

Starred

Dashboards

Explore

Drilldown

Alerting

Alert rules

Contact points

Notification policies

Silences

Active notifications

Recently deleted

Settings

Connections

Administration

Alerting > Alert rules

Q Search...

ctrl+k

+

?



Firing

Normal

Pending

Recovering

Alert

Recording

Ok

No Data

Error

Choose a contact point

Search

Q Search

View as

Grouped

List

State

250 rules

7 firing

149 normal

94 recording

Grafana-managed

Export rules

+ New recording rule

1 firing

3 normal

 |

1m

 |

?

K8s-Cluster-Alerts > node-alerts

State	Name	Health	Summary	Next evaluation	Actions
> Normal	K8s - Node CPU High	ok	Node {{ \$labels.instance }} CPU is above 80%	in a few seconds	View Edit More
> Normal	K8s - Node Memory High	ok	Node {{ \$labels.instance }} Memory is above 85%	in a few seconds	View Edit More
> Normal	K8s - Node Disk High	ok	Node {{ \$labels.instance }} Disk is above 85%	in a few seconds	View Edit More
> Firing for 5m	K8s - Node Down	ok	Node {{ \$labels.instance }} is DOWN	in a few seconds	View Edit More

Gmail

Search mail

?

⚙️

✦

☰

V

✎

📧

☆

🕒

🔍

📄

📁

<

+

2 of 7,772 < > 🗑️

📅 31

💡

✅

👤

+

[FIRING:1] K8s - Node Down K8s-Cluster-Alerts (node-exporter http-metrics 172.31.69.238:9100 node-exporter monitoring monitoring-prometheus-node-exporter-j5qfp monitoring-prometheus-node-exporter warning k8s-cluster) Inbox x

V RxPulse Grafana Alerts <kondojuvinaykumar2004@gmail.com>
to me ▾

15:35 (5 minutes ago) ☆ 😊 ↩ ⋮

Grafana

K8s-Cluster-Alerts › K8s - Node Down

🔥 1 firing instances

Firing	K8s - Node Down	View alert
Summary Node 172.31.69.238:9100 is DOWN		
Description Node 172.31.69.238:9100 is not reachable. Immediate action required.		
Values		

↩ Reply

➜ Forward

😊

>

THANK YOU

RxPulse Application: <http://rxpulse.online>

Github Organization : <https://github.com/orgs/RxPulse/repositories>

Demo- Credentials:

Admin – admin@rxpulse.com

Password: Admin@123

